



Electric Vehicle Charging Infrastructure and Urban Mobility Performance: Evidence from a Tier-II City in India

Sudhakar Madhavedi^{1,2,3*}, Bella Sreekanth Kumar⁴, Ravikanth Regalla^{2,5}, Wong Chee Hoo^{2,3,6}

¹ Department of Business Management, Kshatriya College of Engineering (under JNTU, Hyderabad), 503224 Nizamabad, India

² Faculty of Business and Communications, INTI International University, 71800 Nilai, Malaysia

³ Faculty of Management, Shinawatra University, 12160 Pathun Thani, Thailand

⁴ Department of Business Management, Vijay Rural Engineering College (under JNTU, Hyderabad), 503003 Nizamabad, India

⁵ Department of Sustainable Development, Sphoorthy Engineering College (under JNTU, Hyderabad), 500112 Hyderabad, India

⁶ Department of Economic Sciences, Wekerle Sandor Uzleti Foiskola, 1083 Budapest, Hungary

* Correspondence: Sudhakar Madhavedi (reachfirst@gmail.com)

Received: 12-15-2025

Revised: 01-17-2026

Accepted: 01-23-2026

Citation: S. Madhavedi, B. S. Kumar, R. Regalla, and W. C. Hoo, "Electric vehicle charging infrastructure and urban mobility performance: Evidence from a Tier-II City in India," *Int. J. Transp. Dev. Integr.*, vol. 10, no. 1, pp. 288–308, 2026. <https://doi.org/10.56578/ijtdi100120>.



© 2026 by the author(s). Licensee Acadlore Publishing Services Limited, Hong Kong. This article can be downloaded for free, and reused and quoted with a citation of the original published version, under the CC BY 4.0 license.

Abstract: This study examines how electric vehicle (EV) charging infrastructure influences urban mobility performance in a Tier-II city, using Nizamabad in India as a case study. A structured survey was carried out with EV users—including two-wheeler riders and three-wheeler operators—to explore charging behaviour, accessibility to charging facilities, and the effects on daily travel patterns. The results show that limited charging availability, uneven station distribution, and long waiting times often disrupt trip continuity and reduce operational efficiency. These issues are especially pronounced among commercial operators, who depend on frequent charging to maintain daily service cycles. For them, infrastructure gaps directly affect service reliability and last-mile connectivity. In contrast, users with home charging facilities face far fewer disruptions, pointing to uneven impacts across different user groups. The findings indicate that charging infrastructure functions as a fundamental element of the urban transport system rather than an auxiliary service. A more deliberate alignment of charging stations with activity hubs and transport nodes would improve accessibility and support more stable mobility patterns. These observations underline the importance of coordinated infrastructure planning in facilitating the ongoing transition to electric mobility in emerging urban contexts.

Keywords: Environment; Electric vehicle; Charging infrastructure; Urban mobility performance; Charging dependency; First- and last-mile connectivity; Transport system integration; Accessibility

1 Introduction

Electric vehicles (EVs) are becoming an essential part of India's transition toward sustainable transportation. With rising fuel prices, declining government subsidies, increasing environmental awareness, and advancements in EV technology, adoption rates are growing across both urban and semi-urban regions. Nizamabad, a developing city in Telangana, has also seen a steady rise in electric two-wheelers and three-wheelers, especially among daily commuters. However, the availability of affordable, reliable, and accessible charging infrastructure still lags behind this emerging demand. For small entrepreneurs, EV charging stations offer a unique micro-business opportunity due to low operational costs, flexible investment options, and potential partnerships with government and private companies. The economic feasibility of such a venture, however, depends on several factors, including the demand for charging, users' willingness to pay, preferred locations, charging frequency, the reliability of power supply, and overall perception of EV infrastructure. Understanding these dimensions is crucial to assessing the business potential and profitability of EV charging stations as a micro-enterprise.

The integration of EV charging infrastructure is no longer a peripheral service but a foundational pillar of modern urban transport systems, fundamentally dictating mobility reliability, accessibility, and travel efficiency [1]. In urban landscapes characterized by short- to medium-distance trips, the spatial distribution of these facilities is the primary determinant of whether a journey can be completed without disruption [2]. Inadequate infrastructure leads to systemic failures, such as unplanned travel interruptions, extended waiting times, and inefficient route deviations, which degrade the overall performance of the city's transport network [3]. Consequently, charging infrastructure must be transitioned from a purely economic or entrepreneurial consideration into a core element of comprehensive urban mobility planning and transport development [1].

This study aims to analyze the feasibility of EV charging stations in Nizamabad District by examining user behaviour, satisfaction levels, pricing preferences, and entrepreneurial intentions. It also highlights the challenges currently faced by EV users and identifies the most suitable charging models for the region. By combining user insights with business perspectives, the study provides a comprehensive understanding of how EV charging can develop into a sustainable micro-business opportunity in Nizamabad.

In the specific context of Tier-II Indian cities like Nizamabad, the electrification of the transport sector is predominantly driven by electric two-wheelers and e-rickshaws, which serve as the backbone for daily commuting and last-mile connectivity. These vehicles function as vital feeder services to commercial hubs, markets, and transit stops, yet their operational effectiveness is entirely dependent on the proximity, reliability, and temporal availability of charging nodes. A fragmented charging network forces commercial operators into longer idle times and reduced service frequencies, directly compromising passenger accessibility and the economic viability of green transport. Therefore, a deep understanding of user charging behavior and localized constraints is imperative for evaluating how EV adoption will ultimately impact the efficiency of urban transport systems.

The significance of this study lies in its contribution to understanding the rapidly evolving EV landscape in semi-urban Indian cities, particularly Nizamabad, which is witnessing a gradual shift toward sustainable mobility. With the Government of India and the Telangana State actively promoting EV adoption through various incentives and policies, the demand for reliable charging infrastructure has become increasingly critical. This research offers evidence-based insights into user behaviour, charging needs, and market demand; information that is essential for policymakers, urban planners, and private investors. By identifying the challenges faced by current EV users and examining their willingness to pay, the study provides practical guidance for designing consumer-friendly charging solutions.

This research highlights the potential of EV charging stations as a viable micro-business model, supporting local entrepreneurship and generating employment opportunities. The findings are particularly useful for small and medium investors looking for low-cost, high-potential business ventures in the green technology sector [4]. Finally, this study contributes to sustainable development goals by encouraging cleaner transportation, reducing dependence on fossil fuels, and promoting environmentally responsible business models.

Research Objectives

The present study aims to analyse EV charging infrastructure from a transport development and urban mobility perspective, using Nizamabad District as a case study. The specific objectives are:

1. To examine EV usage patterns, daily travel distances, and charging frequency among electric two-wheeler and three-wheeler users in Nizamabad.
2. To assess how the availability and location of charging facilities influence users' ability to complete daily trips without mobility disruption.
3. To analyse the relationship between travel distance, charging dependency, and waiting time, and their impact on mobility reliability.
4. To identify user groups most dependent on public charging infrastructure, particularly commercial and shared mobility operators.
5. To derive planning implications for integrating EV charging infrastructure with public transport nodes and last-mile connectivity systems.

The scope of this study is limited to assessing the economic feasibility of establishing EV charging stations as a micro-business model in Nizamabad District, Telangana. The research focuses on understanding EV usage patterns among two-wheeler users, three-wheeler auto drivers, and potential early adopters within the district. It also examines the existing charging infrastructure, user satisfaction, charging behaviour, preferred pricing structures, and locational preferences. In addition, the study explores entrepreneurial interest, investment potential, and the type of support expected from government or private entities for setting up EV charging stations. The scope is geographically restricted to Nizamabad and analytically confined to assessing demand, financial feasibility, and behavioural insights derived from structured questionnaire responses. While the study provides valuable insights into the emerging EV ecosystem, it does not include an evaluation of the technical engineering aspects of charging technologies or long-term economic projections beyond the current market environment.

2 Methodology

This section outlines the methodological framework adopted to understand the user preferences and adoption, usage patterns, to assess the economic feasibility of EV charging stations as a micro-business model in Nizamabad District, Telangana. The study is based on a descriptive and exploratory research design, chosen to understand current EV usage patterns, identify infrastructural gaps, and evaluate investment potential in the EV charging ecosystem. The approach enabled systematic measurement of user behaviour, customer satisfaction, and willingness to pay, while the exploratory component helped identify suitable business models and stakeholder expectations in the emerging EV sector. A cross-sectional survey design was used, wherein data collected from respondents at a single point in time, providing a clear snapshot of real-world conditions influencing EV charging demand in the district.

The sample units of this study include users directly associated with EV usage and entrepreneurship in Nizamabad—i.e., EV two-wheeler users, e-auto-rickshaw drivers, business owners, and potential entrepreneurs. Considering the evolving nature of EV adoption in semi-urban cities, purposive sampling was used to ensure that participants could provide meaningful insights related to EV charging demand and business feasibility. A total of 150 respondents were selected, and this sample size was considered appropriate for a preliminary feasibility analysis. These respondents represent a mix of user groups and entrepreneurial segments, enabling a multi-dimensional understanding of the EV charging ecosystem.

Primary data for the study collected using a structured questionnaire designed to address the research objectives. The questionnaire consisted of closed-ended questions, multiple-choice items, and Likert scale statements, respondent demographics, EV usage patterns, satisfaction with existing charging infrastructure, preferred payment methods, and entrepreneurial interest in establishing charging stations. Additional open-ended questions allowed respondents to share qualitative insights on the improvements needed in the charging infrastructure. To support the primary data analysis, secondary data is gathered from government reports and policies, industry publications, and indexed research articles.

To analyze the collected data, responses were coded and processed using Microsoft Excel, and descriptive statistics such as percentages and frequency distributions were applied to summarize demographic information and EV usage patterns. To ensure the credibility of the research instrument, both reliability and validity measures were applied. Content validity was established by consulting experts from academia and the EV technology sector, who reviewed the questionnaire for relevance and clarity. Reliability analysis was conducted using Cronbach's Alpha, which produced values above the acceptable threshold of 0.70, confirming strong internal consistency of the scale items. Ethical considerations were followed throughout the research process: respondents were informed that participation was voluntary, no personally identifiable information was collected, and all responses were treated with strict confidentiality. The data was used solely for academic purposes, in accordance with the ethical standards of social science research.

Despite its methodological strengths, the study acknowledges several limitations. The use of purposive sampling restricts the generalisability of the findings beyond the immediate district. Although the sample size is adequate for a feasibility study, it may not fully capture all emerging categories of EV users, especially given the rapid pace of technological advancements and policy-driven adoption. The reliance on self-reported data may also introduce subjective bias in responses related to charging experiences and entrepreneurial interest. Additionally, future developments in battery technologies, electricity tariffs, and government subsidies may impact the long-term viability of EV charging stations, making some of the findings time-sensitive.

3 Literature Review

The global transition toward sustainable transportation has accelerated the adoption of EVs, driven by climate commitments, technological advancements, and supportive government policies. Numerous studies have highlighted the essential role of charging infrastructure in enabling EV diffusion and ensuring long-term economic viability.

Sierzchula et al. [5] found that the availability of public charging infrastructure is strongly correlated with EV adoption patterns, particularly in developing markets. Similarly, Li et al. [6] emphasised that accessibility and affordability of charging remain the strongest predictors of user satisfaction and continued EV usage. In the Indian context, Bansal and Goyal [7] noted that infrastructural gaps in semi-urban regions limit large-scale adoption, despite rising consumer interest and favourable policy support. Economic feasibility studies have also examined the viability of establishing EV charging stations. Jerome and Udayakumar [8] reported that small-scale, slow-charging models can be economically feasible due to their low operational costs, especially in markets dominated by two- and three-wheelers. Jain et al. [9] observed that investment feasibility improves significantly when user demand, pricing strategies, and station location are optimised.

Researchers have also explored the potential of micro-entrepreneurship in EV infrastructure development. Devanathan and Pillai [10] suggested that micro-business models in the EV sector can succeed when supported by low-interest financing, subsidies, and franchise-based charging models. Jaiswal and Gupta [11] argued that micro-enterprises in Tier-II and Tier-III cities can generate substantial employment opportunities while contributing

to the growth of the EV ecosystem. Several studies also highlight the importance of pricing. Saxena et al. [12] demonstrated that willingness to pay for charging services is shaped by income levels, electricity tariffs, and perceived service reliability. Ghosh and Bhattacharya [13] suggested that flexible payment systems—including prepaid and subscription-based models—can greatly enhance user satisfaction.

Policy-oriented research underscores the role of government support. Gupta et al. [14] observed that India’s Faster Adoption and Manufacturing of EVs (FAME) scheme improved consumer confidence but lacked adequate focus on charging infrastructure development in smaller cities. Kumar et al. [15] recognised that local entrepreneurship could accelerate EV infrastructure expansion when supported through subsidies, training, and equipment access. Turoń et al. [16] reported that EV charging micro-businesses perform well in regions with high demand density, even when charging speeds are low. Meanwhile, Ahmad et al. [17] found that small charging points integrated into commercial and residential zones can generate stable revenue streams in urban–suburban markets.

User charging behaviour has been a key focus in EV research. Wang et al. [18] found that daily travel distance and charging frequency vary significantly across EV types, influencing the preference for home versus public charging. Similarly, Saxena and Shrivastava [19] noted that EV auto-rickshaw drivers depend heavily on public charging stations due to higher daily travel distances, creating strong business potential for micro-level charging facilities. Yilmaz and Krein [20] argued that slow chargers are more suitable for small-scale businesses due to their lower installation costs and reduced grid impact. Likewise, Zhang et al. [21] suggested that battery-swapping stations may be cost-effective in regions dominated by commercial EV users.

Collectively, these studies provide a strong theoretical foundation for the present research, which examines EV charging as a micro-business opportunity in the context of Nizamabad. However, empirical evidence specific to semi-urban Indian cities—particularly related to entrepreneurial interest, charging patterns, and investment feasibility—remains limited. This study aims to fill these gaps.

By looking at the unique transport landscape of semi-urban India, we can see a clear chain reaction: the actual charging conditions like how many stations are available, how long the wait is, and where they are located, directly dictate how people drive and charge their vehicles. These conditions force users to make daily decisions about which routes to take, how often to plug in, and whether to rely on home setups or public hubs. Ultimately, these individual choices ripple outward, affecting the entire city’s mobility performance, leading to either a seamless flow of travel or significant time loss and unreliable trips. For city planners, this means that the success of public transport and last-mile feeder services depends entirely on strategically placing chargers where they can best support these real-world user habits.

4 Data Analysis and Results

4.1 Reliability Analysis

To assess the internal consistency of the measurement instrument, Cronbach’s Alpha (α) was computed for each construct included in the questionnaire. Reliability analysis was performed for five major sections of the survey tool, excluding demographic items and open-ended questions, as these do not contribute to scale reliability.

The results of the reliability test presented in the Table 1 indicates that the questionnaire demonstrates acceptable to high levels of internal consistency across all constructs. The section measuring EV Usage Patterns yielded a Cronbach’s Alpha of $\alpha = 0.78$, which indicates an acceptable level of reliability. Similarly, the construct addressing Satisfaction and Charging Challenges produced a reliability coefficient of $\alpha = 0.81$, suggesting good internal consistency among the items. The Payment, Pricing, and Demand Preferences items recorded a Cronbach’s Alpha of $\alpha = 0.74$, also falling within the acceptable reliability range. The construct measuring Perception toward EV Charging Stations as a Micro-Business demonstrated good reliability with $\alpha = 0.83$. Finally, the Entrepreneurial Interest and Support Expectations section achieved a high reliability value of $\alpha = 0.86$, representing strong internal consistency and indicating that the items within this construct reliably capture the underlying concept. These results confirm that the questionnaire items consistently measure the intended constructs related to EV charging behavior, satisfaction levels, entrepreneurial inclination, and perceptions toward EV charging stations as a micro-business model in Nizamabad.

Table 1. Reliability analysis

S. No	Section/Construct	Number of Items	Cronbach’s Alpha (α)	Reliability Level
1	Electric Vehicle (EV) Usage Patterns	6	0.78	Acceptable
2	Satisfaction and Charging Challenges	3	0.81	Good
3	Payment, Pricing, and Demand Preferences	4	0.74	Acceptable
4	Perception Toward EV Charging Stations	3	0.83	Good
5	Entrepreneurial Interest and Support	4	0.86	High

4.2 Demographic Study

Table 2 represents the demographic study that consists of frequency count and percentages of all responses to the demographic variables of 150 respondents from Nizamabad District, Telangana.

Table 2. Demographic study

Demographic Variable	Category	Frequency	Percentage (%)
Age Group	Below 20	8	5.3
	21–30	50	33.3
	31–40	42	28.0
	41–50	28	18.7
	Above 50	22	14.7
Gender	Male	101	67.3
	Female	43	28.7
	Other	6	4.0
Occupation	Electric vehicles (EV) 2-wheeler user	63	42.0
	EV 3-wheeler (auto) driver	41	27.3
	Business owner	16	10.7
	Potential entrepreneur	13	8.7
	Other	17	11.3
Monthly Income	Below international normalized ratio (INR) 10,000	29	19.3
	INR 10,001–INR 20,000	41	27.3
	INR 20,001–INR 40,000	52	34.7
	INR 40,001–INR 60,000	16	10.7
	Above INR 60,000	12	8.0
Area of Residence	Nizamabad Urban	72	48.0
	Nizamabad Rural	60	40.0
	Other	18	12.0

4.2.1 Age group and gender

A larger proportion of respondents (61.3%) fall between 21 and 40 years of age, indicating that EV adoption and interest in related business opportunities are largely concentrated among young to middle-aged adults who are more tech-savvy and open to new mobility solutions. The presence of respondents aged 41–50 and above 50 (33.4%) further suggests growing EV interest among older adults, reflecting a wider age appeal. The gender distribution shows male dominance (67.3%), which is common in India's mobility sector; however, the notable participation of women (28.7%) and respondents from other gender identities (4%) highlights increasing inclusivity and the potential for targeted outreach initiatives to further engage women and diverse groups in EV usage and entrepreneurship.

4.2.2 Occupation profile

The majority of respondents are EV two-wheeler users (42%) and three-wheeler (auto) drivers (27.3%), highlighting the dominance of these vehicle types in Nizamabad's local EV market. Business owners (10.7%) and potential entrepreneurs (8.7%) together account for nearly one-fifth of the sample, reflecting a strong entrepreneurial presence capable of supporting or expanding EV infrastructure and services. This diverse mix suggests significant potential for micro-businesses focused on EV charging stations, maintenance, and related services, aligning well with local demand and livelihood opportunities.

4.2.3 Income distribution and area of residence

Most respondents fall within the international normalized ratio (INR) 10,000 to INR 40,000 monthly income range (61.9%), indicating a predominantly lower-middle to middle-income demographic. This suggests that affordability is a key factor influencing both EV adoption and willingness to pay for charging services. Consequently, infrastructure and pricing models should prioritise cost-effectiveness to ensure accessibility for this segment. The nearly even distribution between urban (48%) and rural (40%) residents indicates that EV adoption and related business opportunities extend across diverse geographies, highlighting the need for inclusive infrastructure planning that supports both urban convenience and rural reach to maximize impact.

The demographic profile reflects a young and moderately diverse user base, with a substantial proportion well-positioned for EV adoption and entrepreneurial ventures. The strong presence of two- and three-wheelers aligns with national trends, highlighting these vehicles as key areas for infrastructure investment. The income and urban–rural diversity underscore the need for flexible, affordable, and accessible charging solutions. Furthermore, the notable

entrepreneurial interest (19.4% combined business owners and potential entrepreneurs) suggests readiness to engage with the EV ecosystem, particularly through micro-businesses such as localized charging stations, which could be supported by government subsidies and training programs.

4.3 Analysis on Electric Vehicle Usage Pattern

4.3.1 Type of electric vehicle used

According to the Figure 1, electric two-wheelers dominate the EV market, accounting for nearly half of the total sample (48%), reflecting that the current EV transition is largely driven by personal transportation needs. Electric scooters and motorcycles offer an attractive balance of affordability and utility, making them the primary substitute for petrol-based daily commuting. Electric three-wheelers account for 23.3% of respondents, representing a significant share that supports public transport and last-mile connectivity. Unlike two-wheelers, this segment primarily consists of mini commercial vehicles, and their presence underscores the economic advantages of EVs, such as lower running costs, government subsidies, and higher returns on investment. In contrast, electric four-wheelers make up only 8.7% of users, highlighting low adoption due to the “premium barrier.” While two- and three-wheelers have achieved mass-market appeal, electric cars remain a niche segment because of higher upfront costs and reliance on robust charging infrastructure, which is still developing.

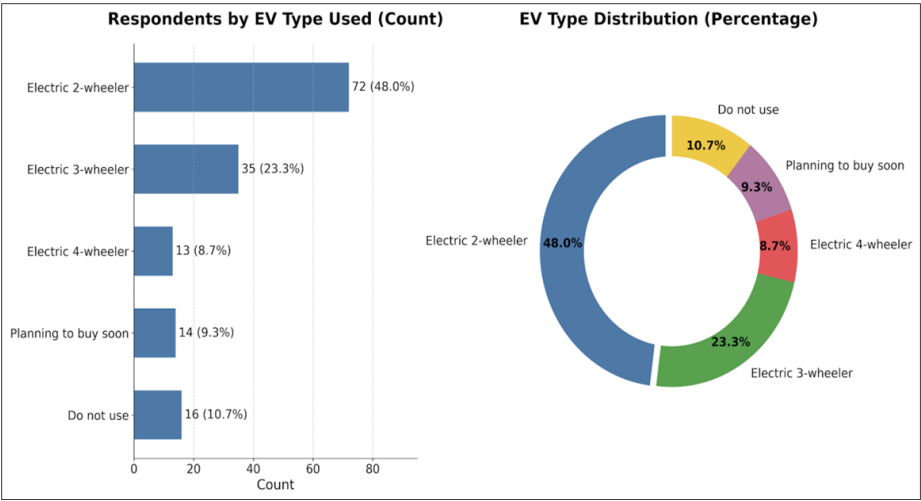


Figure 1. Distribution of electric vehicle (EV) types

The “Planning to Buy” category (9.3%) represents the immediate potential market, nearly equal in size to the current four-wheeler user base, indicating that the market is far from stagnant. If these potential buyers convert, the total EV user base could grow by roughly 10% in the near term. Only 10.7% of respondents fall into the “Do Not Use” category, a figure much lower than typically observed in general population surveys. This low proportion suggests that the survey targeted an EV-aware ecosystem or that EV technology has already reached a high level of normalization within this demographic. The EV revolution in this group is clearly driven not by luxury cars but by affordable, high-utility vehicles such as two- and three-wheelers. The combined 71.3% share of these segments underscores that cost savings and practical utility are the primary factors driving electrification in this market.

4.3.2 Duration of electric vehicle usage

According to the data shown in Figure 2, the EV market in the study is transitioning from an “early adopter” phase to a stage of market growth. The largest single segment of respondents (27.3%) has used EVs for 1–2 years, while a substantial 24.0% have been using them for more than two years. Combined, users with over one year of experience account for 51.3% of the sample. This is a key finding, as it indicates that the feedback collected reflects sustained, long-term usage rather than initial novelty. These experienced users have likely gone through the full cycle of vehicle ownership, including battery aging, maintenance requirements, and seasonal performance variations.

While the established user base is strong, the influx of new users remains steady. Respondents with less than six months of experience (18.7%) and those with six months to one year (19.3%) together make up 38.0% of the sample. The nearly equal split between these cohorts indicates a consistent rate of new vehicle adoption over the past year, rather than a sudden, volatile surge. The 10.7% of respondents classified as “not an EV user” aligns with the non-user percentage reported earlier, providing internal consistency and validating the data quality across survey questions. Overall, the EV market comprises both repeat users and new adopters, reflecting a sustainable balance of retention and growth rather than a short-term bubble.

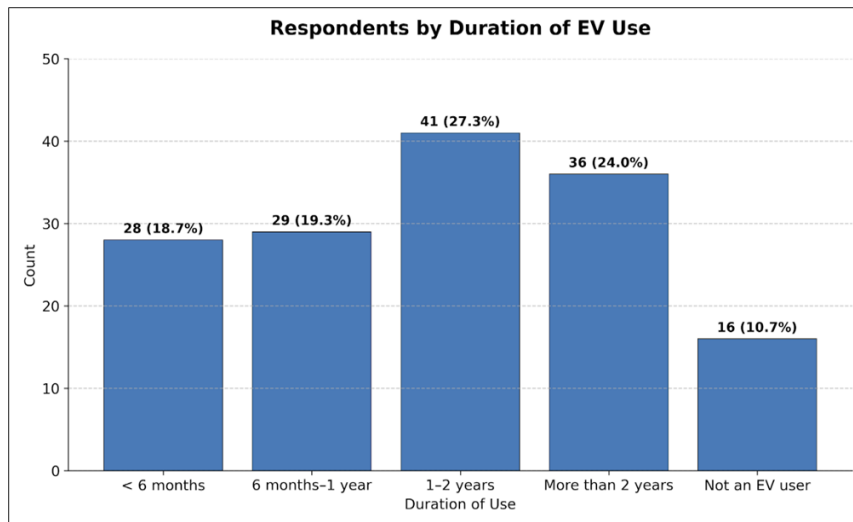


Figure 2. Duration of electric vehicle (EV) usage

4.3.3 Daily average travelling distance

Figure 3 shows daily travelling distances by EV users; most of the users travel between 20 and 40 km daily, with 38.7% (58 respondents) falling into this category. This distance aligns with typical urban travel in Tier-II cities, such as a 10–20 km round trip for work or business, indicating that current EV batteries are sufficient for the majority of users' daily needs, often requiring only a single charge per day or every other day. The next largest group, traveling less than 20 km daily, accounts for 20.0% of users. Their charging needs are minimal, and the operational cost savings compared to petrol vehicles over time are substantial. About 21.3% (32 respondents) travel between 40 and 60 km daily, representing users with longer commutes or those using their EVs for commercial purposes. While still within the range of most modern EVs, this group would need a disciplined daily charging routine. Only 9.3% (14 respondents) travel more than 60 km daily. This small segment likely perceives current EV options as less viable for long distances—due to range anxiety or limited mid-day public charging—or reflects the reality that few people have such long daily commutes. This group is most prone to experiencing charging-related anxiety.

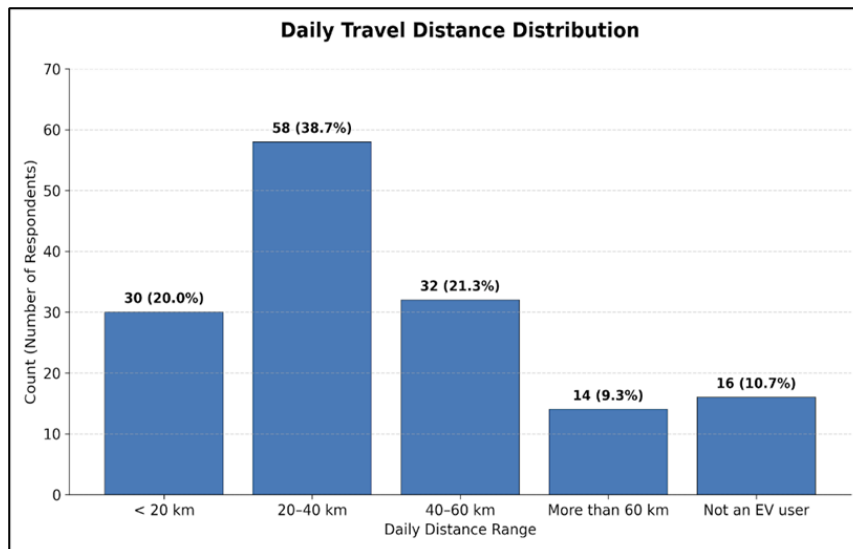


Figure 3. Average daily travel distances

The cumulative data indicate that 80% of active EV users—combining the <20 km, 20–40 km, and 40–60 km groups—travel 60 km or less daily. This finding strongly supports the practical viability of current EV technology in urban and semi-urban settings, demonstrating that for the vast majority of users, vehicle range is not a limiting factor for everyday commuting or routine travel.

4.4 Analysis on User Charging Preferences

4.4.1 Electric vehicle charging patterns

According to Figure 4, the most common charging behavior, reported by 42.7% of respondents (64 individuals), is once-daily charging, reflecting a routine “charge-at-home overnight” habit. This behavior is similar to charging a mobile phone—users plug in at the end of the day regardless of the remaining battery level to ensure a full range for the next day, effectively minimizing range anxiety. A smaller but significant segment (12.7%, 19 respondents) charges twice daily, likely corresponding to commercial users, such as auto or three-wheeler drivers, or high-mileage commuters for whom a single overnight charge is insufficient, highlighting the need for accessible public charging infrastructure or daytime battery-swapping stations. Interestingly, 26.7% of users (40 respondents) charge only once every two to three days, indicating that modern EV batteries often exceed daily travel needs. These users are likely short-distance commuters (<20 km/day) who treat their EV more like a petrol vehicle—refueling only when necessary—rather than a mobile device. A minority of 7.3% charges once weekly, likely representing ultra-short-distance users or those using their EV as a secondary vehicle for occasional errands.

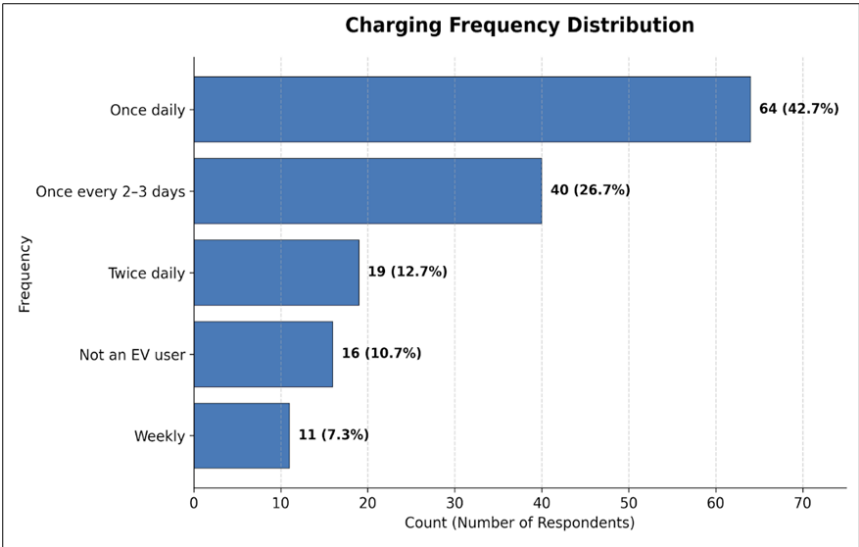


Figure 4. Frequency of electric vehicle (EV) charging

The data reveals a clear split in charging behavior. While the majority of users (55.4%, combining “once daily” and “twice daily”) falls under high-frequency charging category, a significant minority (34%, combining “every 2–3 days” and “weekly”) follows a low-frequency charging schedule. This indicates that while accessible public charging infrastructure is crucial for the high-intensity group, home charging solutions are adequate for most users who charge every two to three days or weekly.

Our research shows a direct link between how far people drive and how much they rely on public chargers, creating a clear divide in the EV experience. Commercial operators and long-distance drivers, particularly those in electric three-wheelers, are the most dependent on public stations and, as a result, face the highest risk of delays or service gaps when chargers are occupied or broken. On the other hand, people who commute shorter distances on electric two-wheelers usually just charge at home, avoiding these interruptions altogether. This gap proves that a lack of public charging doesn’t affect everyone equally—it hits our commercial and shared transport services the hardest, which ultimately makes our city’s overall transport system less reliable and harder for everyone to use.

4.4.2 Preferred charging location

According to Figure 5, about 40.0% of EV user respondents prefer home charging is the primary choice. This aligns closely with the “once daily” charging pattern and confirms that the dominant ownership model in this ecosystem is “charge while you sleep.” This convenience is a key adoption driver, eliminating the need to visit a fuel station for nearly half the users. Interestingly, 33.3% (50 respondents) depend on public charging stations as their main source. This segment likely includes three-wheeler commercial users and those without dedicated home charging facilities, such as apartment residents, highlighting that public infrastructure is not merely a backup but a primary enabler for a significant portion of the EV community.

Only 4.7% use workplace charging as their primary option, representing a largely untapped opportunity. Low adoption may be due to limited charging infrastructure at office premises or a lack of incentives for employers to provide such facilities. Roadside informal shops cater to 8.7% of users, illustrating the grassroots nature of the

ecosystem—general stores or automobile shops often fill gaps for two- and three-wheelers where formal infrastructure is absent. Overall, the charging ecosystem is a hybrid model, dominated by home charging (40%) but strongly supported by public stations (33.3%). Future infrastructure investment should focus on two areas: residential solutions, such as retrofitting apartments for home charging, and dense public networks to support the one-third of users who cannot charge at home.

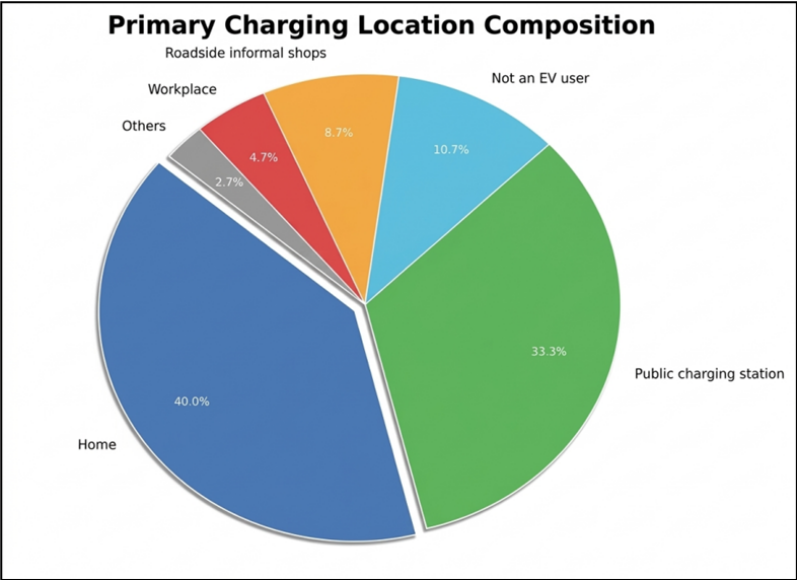


Figure 5. Preferred electric vehicle (EV) charging location

4.4.3 Use preference on charging time

The analysis of preferred charging times provides valuable insights into daily mobility patterns and user behavior. As per Figure 6, Evening hours are the most favored, accounting for 41.3% of responses, likely reflecting users returning home from work or daily activities and charging their EVs overnight for the next day. This highlights the need for adequate charging infrastructure during peak evening hours. Morning charging is the second most preferred, chosen by 22.7% of respondents, suggesting that a significant portion of users prefer to charge before starting their daily commute, which can help avoid battery overcharging and optimize usage. Afternoon and late-night charging each account for 18% of responses, showing moderate preference. Afternoon charging may coincide with commercial activities, longer commutes, or workplace needs, while night charging may suit users with shift-based schedules or those seeking less congestion at stations.

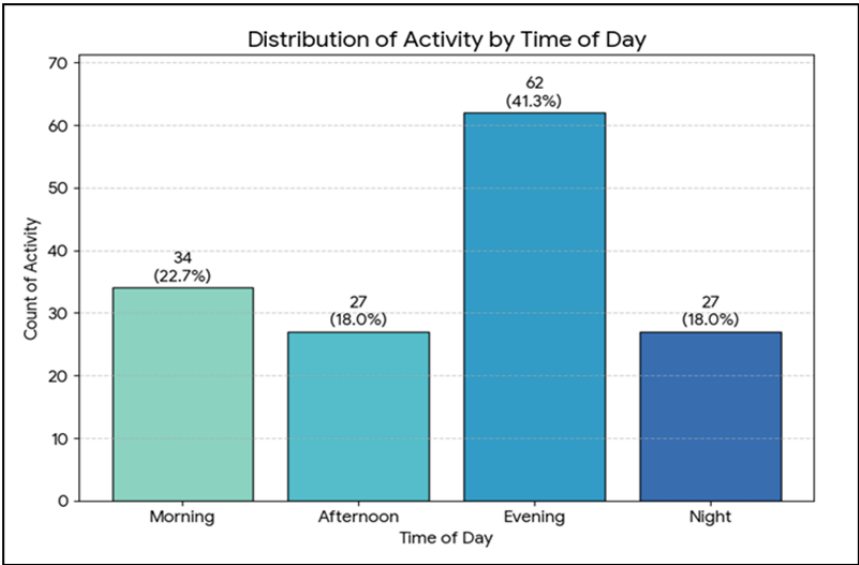


Figure 6. Preferred time for charging electric vehicles (EVs)

These findings are crucial for planning load distribution, optimizing charging infrastructure, and designing time-based tariff models. Understanding temporal charging patterns can help policymakers and service providers reduce congestion, enhance user satisfaction, and improve operational efficiency.

4.4.4 Average charging duration

Figure 7 highlights most common charging duration is 1–2 hours, reported by 44.7% of respondents, indicating that nearly half of users find a mid-range charging time sufficient, likely corresponding to “top-up” charging at public stations or standard home/work charging. The next most popular duration is 2–3 hours, representing 20.0% (30 individuals). Short sessions of less than 1 hour account for 14.7%, likely representing fast-charging users or those needing only a quick range extension. Only 10.0% of users charge for more than 3 hours, suggesting that deep, long-duration charging—typically associated with slow AC chargers for fully depleted batteries—is the least common daily habit. A further 10.7% of respondents identified as non-EV users. Overall, the data reveals a strong preference for moderate charging sessions: combining the top two categories, 64.7% of users charge for 1–3 hours. This insight is valuable for infrastructure planning, indicating that chargers optimized for 1–3 hour turnover would serve the majority of the user base most effectively.

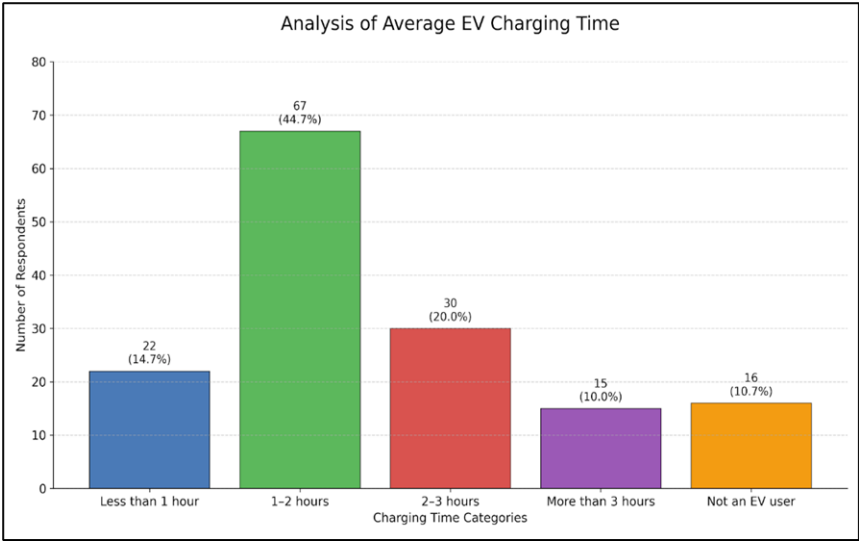


Figure 7. Distribution of electric vehicle (EV) charging time

When you’re racing against the clock—whether you’re an e-rickshaw driver trying to hit a daily target or a commuter heading to work—long wait times at a charging station feel like much more than just a minor annoyance; they represent a complete breakdown in the city’s flow. For the people who keep our streets moving, every extra minute spent waiting for a plug is a minute they aren’t earning or moving, which ultimately thins out the available transport for everyone else and turns what should be a simple charge into a major roadblock for the whole city’s efficiency.

4.4.5 User willingness to pay per charging

The analysis highlights user preferences for pricing, offering valuable guidance for tariff and revenue planning for EV charging infrastructure. As per Figure 8, the majority of users (32.7%) prefer a moderate pricing range of Rs. 31–40, indicating a perception of fair balance between affordability and value compared to non-renewable energy. Pricing between Rs. 20–30 and Rs. 41–60, each preferred by 27.3% of respondents, shows comparable acceptance levels. The lower range reflects price-sensitive users, likely influenced by lower incomes or shorter daily travel distances, while the higher range indicates a segment willing to pay more for faster charging, better service quality, or convenient locations.

A smaller proportion, 12.7% of respondents, are willing to pay above Rs. 60, suggesting that premium pricing appeals to only a limited segment. Collectively, these findings suggest that operators should adopt differential pricing models catering to diverse user segments—from budget-conscious daily commuters to premium users seeking advanced services. The predominance of mid-range preferences emphasizes the need for competitive pricing to ensure broad adoption and regular utilization of charging infrastructure. These insights are critical for designing financially viable solutions while promoting wider EV adoption in the region.

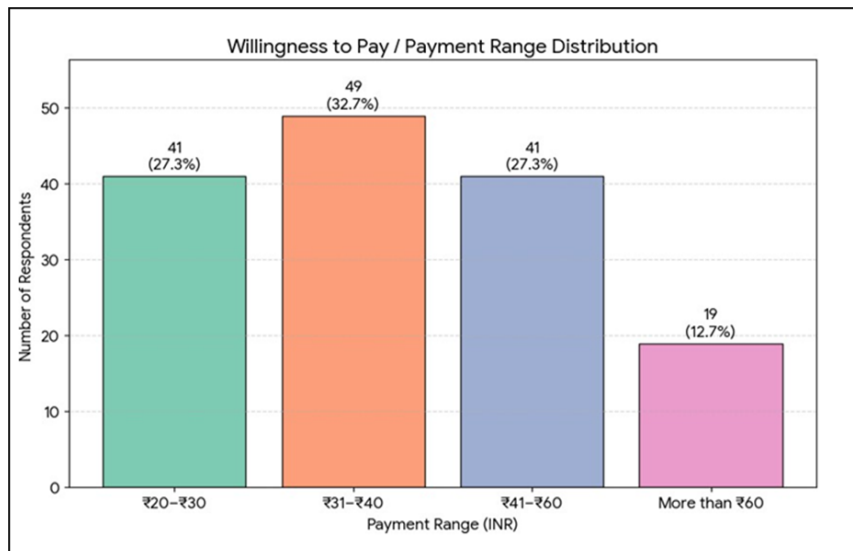


Figure 8. User willingness to pay for electric vehicle (EV) charging service

4.4.6 Preferred payment mode

The analysis of preferred payment methods offers insights into user convenience and expectations. Figure 9 reveals that the Unified Payments Interface (UPI) emerges as the most favored option, chosen by 45.3% of respondents, reflecting the widespread adoption of mobile-based digital payments in urban and semi-urban India due to their ease of use, instant transactions, and reliability. This dominance indicates that EV charging operators should integrate seamless digital payment systems to enhance user experience and transaction efficiency. Cash payments, preferred by 29.3% of respondents, remain the second most common method, highlighting that a significant segment still relies on traditional transactions—likely due to demographic factors, digital literacy gaps, or a preference for tangible payments.

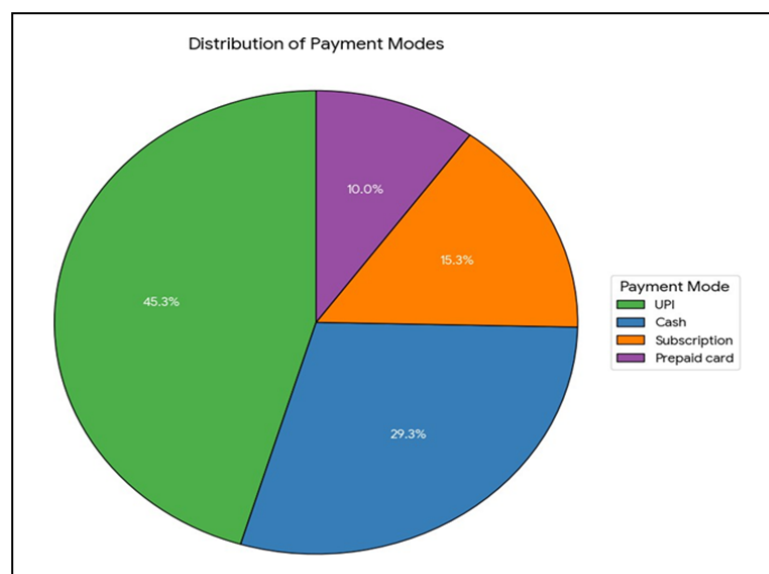


Figure 9. Preferred mode of payment for electric vehicle (EV) charging service

Consequently, retaining cash options is essential to cater to diverse users, especially first-time EV adopters or those from less digitalized backgrounds. Moderate interest is seen in subscription-based payments (15.3%), suggesting a willingness among some users—such as frequent commuters or fleet operators—to adopt periodic or membership models that could enhance loyalty and provide predictable revenue for operators. Prepaid cards, preferred by only 10% of respondents are the least favored, likely due to inconvenience and limited flexibility compared with real-time digital alternatives like UPI. Overall, the findings underscore a clear shift toward digital, real-time, and user-friendly payment solutions, with UPI firmly established as the dominant mode.

4.5 Analysis of User Perception

4.5.1 Satisfaction with charging facilities

The data presented in Figure 10 reveals a mixed sentiment among respondents regarding the charging facilities. The largest single group, Neutral, comprises 30.7% (46 respondents), indicating that a significant portion of users are neither particularly satisfied nor dissatisfied. This “fence-sitting” group represents a critical opportunity, as targeted improvements could convert them into satisfied users. Combining “Very Satisfied” (10.7%) and “Satisfied” (27.3%) shows that 38.0% of users have a positive view, while “Very Dissatisfied” (10.7%) and “Dissatisfied” (20.7%) together account for 31.4% expressing negative sentiment.

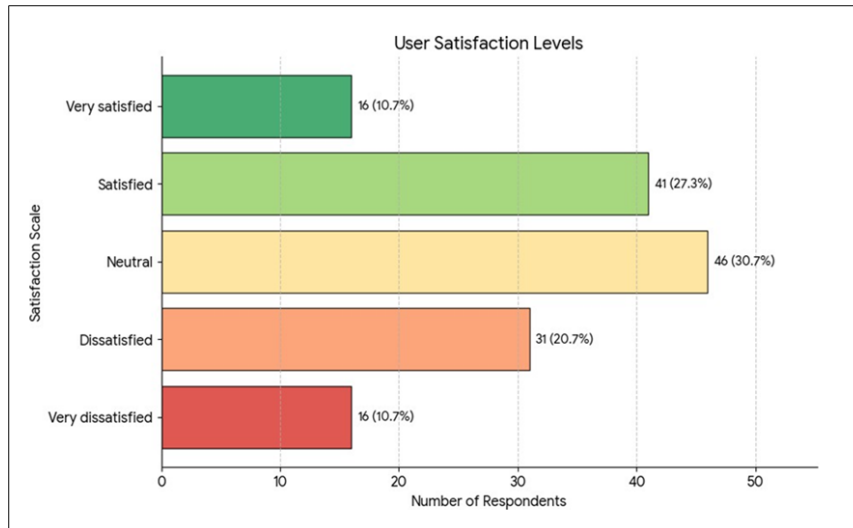


Figure 10. User satisfaction with electric vehicle (EV) charging facilities

Although positive sentiment slightly outweighs negative sentiment by 6.6%, the large neutral block heavily anchors overall perception. This balance suggests a polarized experience for roughly 20% of the user base, where highly loyal advocates are mirrored by strong detractors. While more users are satisfied than dissatisfied, the combined neutral and dissatisfied responses—exceeding 50%—indicate that the product or service meets basic needs but fails to delight most users. Efforts should therefore focus on addressing the pain points of the dissatisfied group and identifying “wow” factors to engage and convert the neutral users.

4.5.2 Challenges faced by electric vehicle users

The data derived from a “multi-select” option presented in Figure 11, where percentages exceed 100% (155.2%), indicates that many respondents face multiple simultaneous challenges. The findings highlight that infrastructure availability is a far more pressing concern than cost. The lack of charging stations is the top issue, cited by 51.3% of respondents—more than double the frequency of the next most common challenge—indicating that range anxiety is primarily driven by the absence of charging points rather than charger performance. Unreliable power supply (25.3%) ranks second, and when combined with poor location accessibility (19.3%), it shows that even where stations exist, they are often offline, broken, or difficult to reach. About 22.7% of users reported long waiting times, while 15.3% cited slow charging as a concern.

Surprisingly, high charging costs (12.0%) were among the least cited issues, suggesting that current EV users are relatively price-insensitive but highly sensitive to convenience and availability—they are willing to pay if the service is reliable and accessible. Only 9.3% of respondents reported no challenges, implying that over 90% face some form of difficulty. To improve user satisfaction, priorities should focus on expanding network coverage and enhancing reliability rather than lowering prices or marginally increasing charging speeds.

4.5.3 Need for more charging stations

The study on willingness to use EV charging stations reveals a positive outlook toward the adoption of charging infrastructure. According to Figure 12, a large majority of respondents (81.3%) expressed clear willingness to use EV charging facilities, reflecting growing confidence in EVs’ reliability, convenience, and the necessity of accessible charging networks. This high level of willingness serves as a strong indicator of future demand and supports further investment in EV charging infrastructure. Only 8% of respondents stated they would not use charging stations, likely including non-EV users, those with limited exposure to EV technology, or individuals facing barriers such as cost, accessibility, or perceived utility.

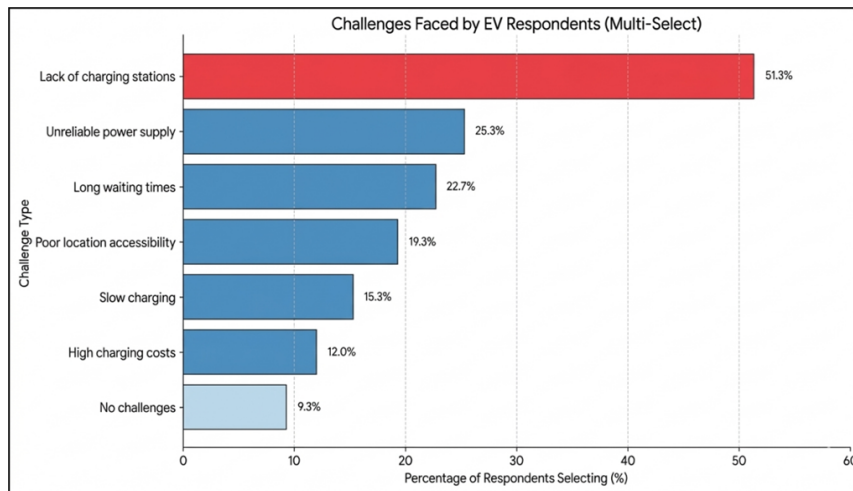


Figure 11. Challenges faced by electric vehicle (EV) users

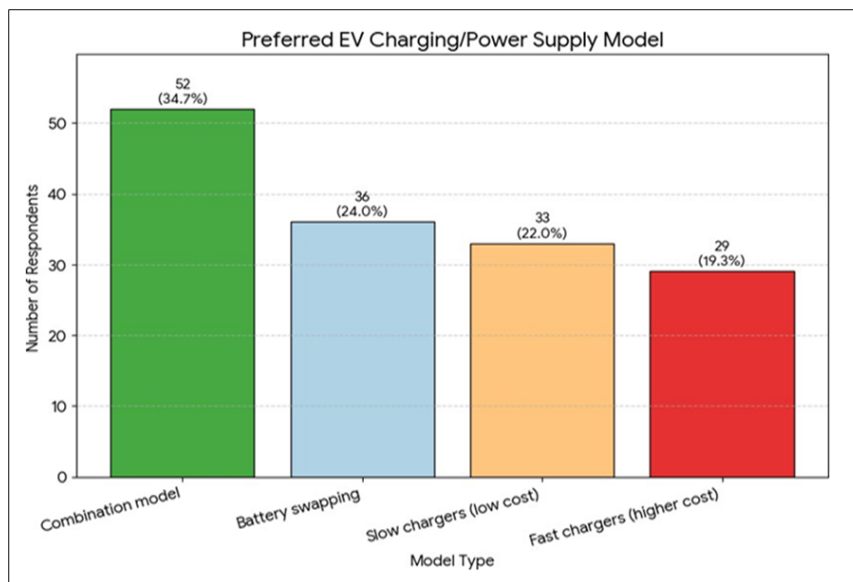


Figure 12. Preferred electric vehicle (EV) charging model

Meanwhile, 10.7% were unsure, suggesting unfamiliarity with EV usage or charging processes. This group represents a key target for awareness campaigns, educational initiatives, and demonstration projects to build confidence and promote adoption. Overall, the findings point to a highly encouraging environment for expanding EV charging networks, with strong potential for rapid growth in EV-related services.

4.5.4 Likelihood to recommend new charging stations

The likelihood of recommending EV charging stations offers valuable insights into user satisfaction, perceived service value, and the potential for word-of-mouth promotion, which is critical for accelerating EV adoption. The results presented in Figure 13 reflect a strongly positive outlook, with 36% of respondents indicating they are “Likely” and 31.3% “Very Likely” to recommend new facilities, together accounting for 67.3% of the total responses. This demonstrates strong user confidence in EV charging infrastructure and its perceived usefulness. A moderate segment (16%) expressed a neutral stance, neither endorsing nor discouraging use, suggesting satisfactory but not exceptional experiences. Their neutrality may stem from limited exposure, evolving expectations, or service quality gaps, and targeted improvements in charging speed, convenience, and pricing could convert them into positive promoters.

Conversely, 9.3% of respondents were “Unlikely” and 7.3% “Very Unlikely” to recommend stations, representing 16.6% of the sample. This group may reflect dissatisfaction related to cost, accessibility, or reliability. Understanding these concerns is essential for service providers and policymakers to refine operations, improve user experience, and address barriers. Overall, the findings indicate a predominantly optimistic outlook toward recommending EV charging infrastructure.

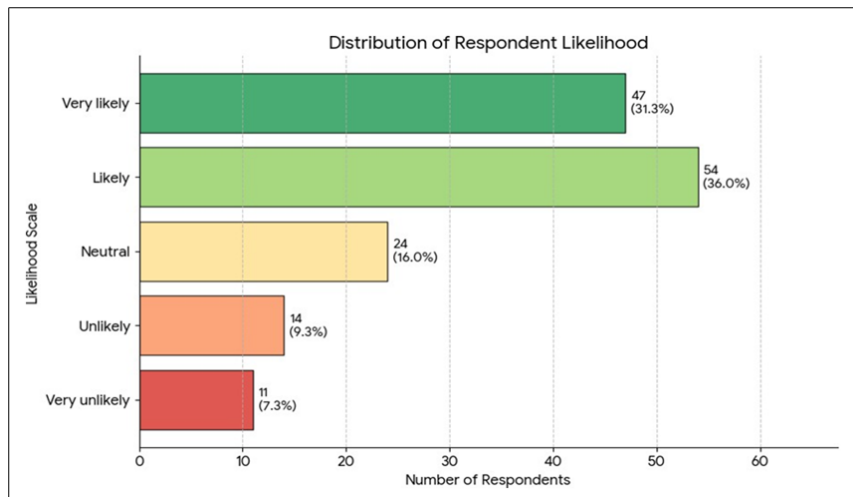


Figure 13. Likelihood of recommending electric vehicle (EV) charging stations

4.6 Entrepreneurial Opportunities

4.6.1 Preferred charging station model

The analysis of preferences for EV charging business models highlights varying user priorities and potential business approaches. As per Figure 14, the most favored option is the combination model, integrating multiple charging solutions such as slow charging, fast charging, and possibly battery swapping, chosen by 34.7% of respondents. This preference underscores the importance of flexibility and adaptability in EV infrastructure to accommodate diverse vehicle types, charging speeds, and usage scenarios. Battery swapping ranks second at 24%, appealing particularly to commercial users, fleet operators, and high-mileage vehicles that prioritize efficiency and rapid turnaround.

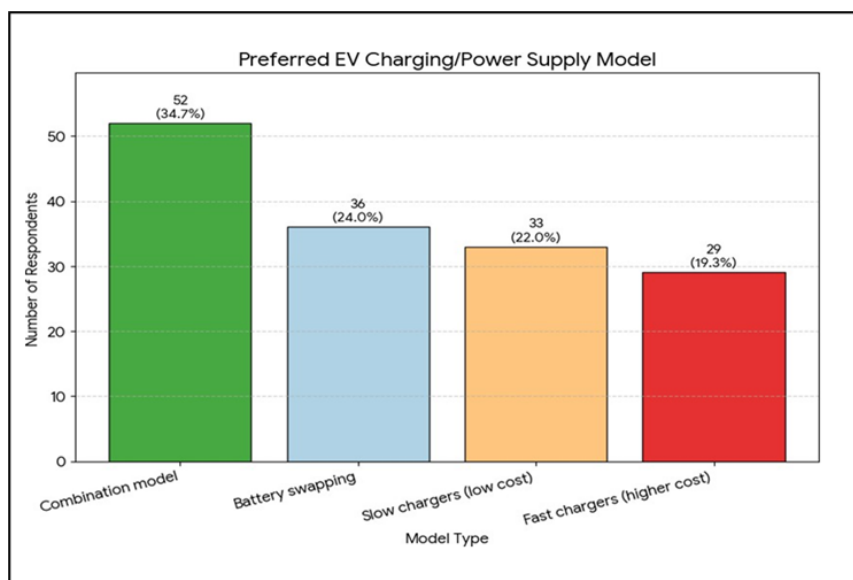


Figure 14. Preferred charging station model

Slow chargers, preferred by 22%, reflect a substantial segment seeking cost-effective solutions suitable for routine overnight charging and residential deployment. Fast chargers, despite their technological advantages, received the least preference at 19.3%, likely due to higher installation and operational costs and the perception that they are unnecessary for typical daily commutes. Overall, the findings indicate a strong demand for versatile, multi-solution charging ecosystems, with cost-effectiveness and operational efficiency being key factors influencing user choices.

4.6.2 Entrepreneurial interest

The respondent comprises of business owners and potential entrepreneurs demonstrated a largely positive outlook toward establishing EV charging businesses. According to Figure 15, about 58.6% expressed clear interest in

starting such a venture, indicating strong entrepreneurial potential and recognition of the commercial viability, growth prospects, and future relevance of EV infrastructure. Their willingness is likely influenced by factors such as increasing government support, rising EV adoption rates, and perceived long-term profitability.

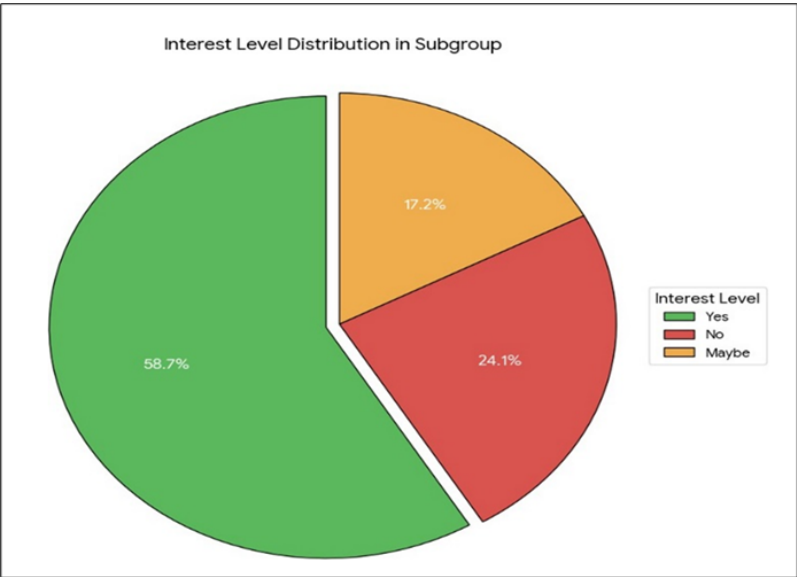


Figure 15. Entrepreneurial interest among business owners and potential entrepreneurs

Conversely, 24.1% indicated no interest, reflecting a minority who may be deterred by perceived risks, financial constraints, limited technical knowledge, or uncertainty about market stability. This highlights the need for awareness programs, feasibility demonstrations, and financial guidance to reduce barriers. Additionally, 17.2% were unsure about establishing an EV charging enterprise, possibly due to limited understanding of operational requirements, unclear revenue expectations, or lack of exposure to successful business models. Overall, the findings demonstrate a strong positive orientation toward EV charging entrepreneurship within this subgroup, with more than half showing clear interest in pursuing such ventures.

4.6.3 Preferred business model

The analysis offers valuable insights into the entrepreneurial models considered most viable to enter into EV charging business. The findings shown in Figure 16 show that the Own & Operate model is the most preferred, selected by 34.5% of respondents, reflecting a strong desire for full ownership control, operational flexibility, and the potential for higher returns. This choice indicates confidence in managing the day-to-day operations of EV charging infrastructure independently. The Shared Revenue model ranks second at 24.1%, appealing to those who favor risk-sharing and collaborative investment structures, as it can reduce financial burden and provide operational support from experienced partners or service providers.

Franchise and partnership models, each preferred by 20.7%, show moderate interest. The franchise model, in particular, attracts individuals seeking structured support, brand recognition, and standardized operations offered by established EV companies. Overall, these findings highlight the need for diverse business entry pathways within the EV charging ecosystem to accommodate entrepreneurs with varying financial capacities, risk tolerance, and operational readiness.

4.6.4 Preferred investment outlay

The preferred investment range for entering the EV charging business is presented in Figure 17, which is found to be balanced and realistic. The most favored bracket, Rs. 50,000 to Rs. 100,000, was chosen by 34.5% of respondents, reflecting a preference for low-entry-cost business models due to limited financial capacity, risk sensitivity, or a desire to start with minimal infrastructure and operational expenses. The second most preferred range, Rs. 100,000 to 200,000, was selected by 31% of respondents, indicating a willingness to commit moderate investment, potentially for more robust charging setups or small-scale fast-charging options. The near-equal preference between these two categories highlights a cautious yet optimistic investment mindset among the subgroup.

Higher investment ranges, Rs. 200,000 to 300,000 and above Rs. 300,000, together account for 17.2% of responses, representing individuals willing to commit significant capital, likely targeting advanced solutions such as fast chargers or hybrid models integrating multiple charging options. Their readiness to invest more reflects confidence in the long-term growth of the EV sector and the potential for higher returns from premium services.

These insights are essential for designing tailored financial support programs, government subsidies, and investment models that align with the varying capital capacities of prospective EV charging entrepreneurs.

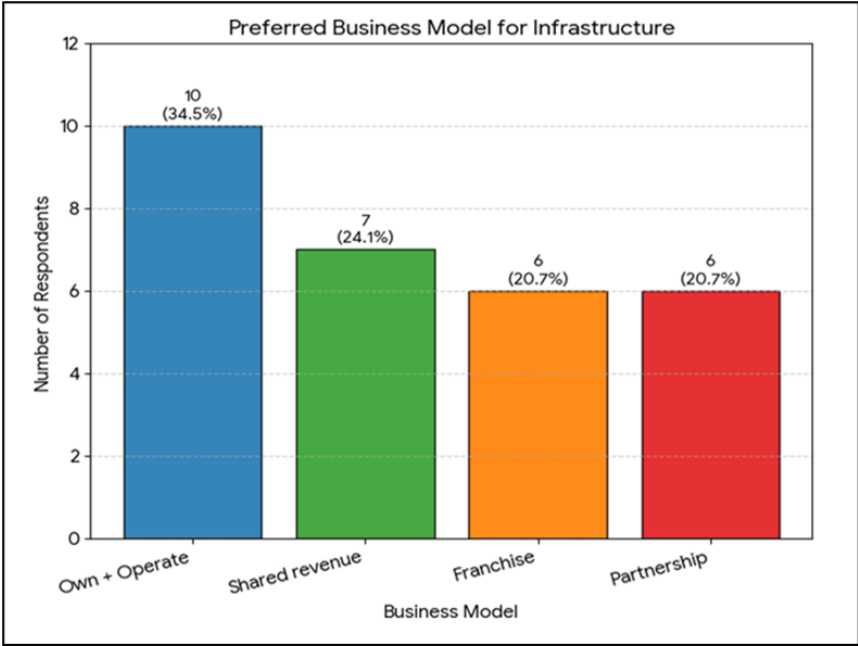


Figure 16. Preferred business model for establishment of EV charging station

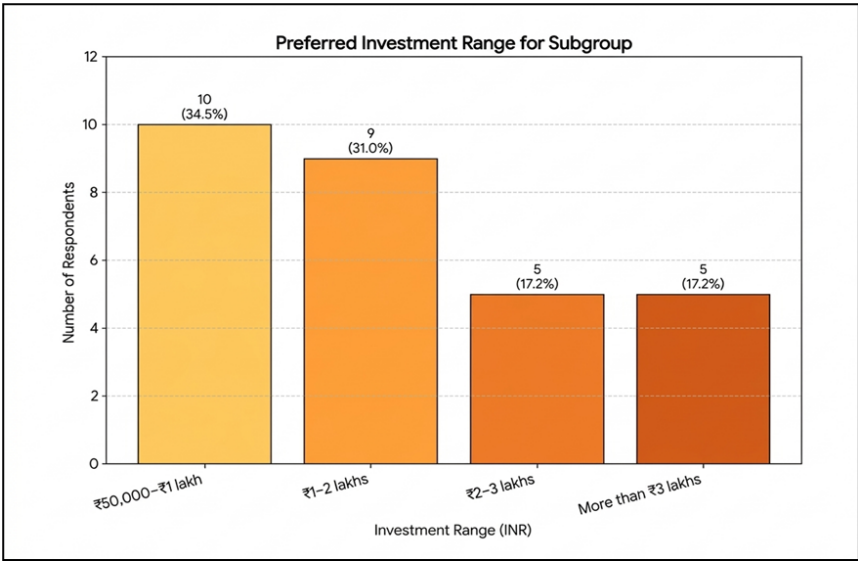


Figure 17. Preferred investment range for establishment of EV charging stations

4.6.5 Expected support eco-system

The support requirements for starting an EV charging business highlight the key enablers that prospective entrepreneurs consider essential for entry. The findings presented in Figure 18 reveal that financial assistance in the form of government subsidies is the most sought-after support, cited by 55.2% of respondents, reflecting the critical role of capital incentives in mitigating investment risks and encouraging wider participation. Training and technical assistance were expected by 41.4% of respondents, underscoring the need for specific technical knowledge in equipment handling, electrical safety, software management, and maintenance protocols. This demand points to the importance of capacity-building initiatives, such as workshops and hands-on training programs, to enhance the competence and confidence of new entrants.

Accessible bank loans were preferred by 34.5% of respondents, indicating that flexible financing options, simplified application procedures, and risk-sharing mechanisms are essential to support small- and medium-scale

investors in the EV sector. Additionally, about 24.1% of respondents sought support for equipment supply and land/space availability, reflecting infrastructural concerns. Access to affordable chargers, related hardware, and suitable locations is considered a vital component for establishing operationally viable charging stations.

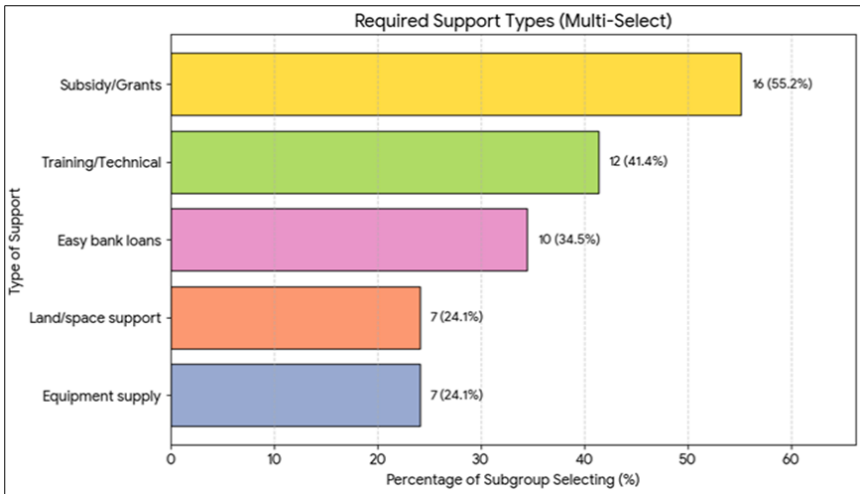


Figure 18. Support system required for EV charging station business

5 Findings and Discussion

The demographic profile shows that EV adoption in Nizamabad is led by younger and middle-aged users, with males forming the majority [22]. Two-wheelers and three-wheeler auto drivers dominate usage, reflecting national trends where younger consumers and utility-focused segments drive EV uptake due to cost and efficiency advantages. These results are in consistent with Ali and Gupta [23]; Singh and Das [24]; Dutta and Mandal [25] and International Energy Agency (IEA) [26].

Affordability drives market composition, with 2-wheelers and 3-wheelers dominating, while 4-wheeler adoption remains low, reflecting higher cost barriers as observed in emerging-market EV studies [27, 28]. The duration-of-use data show that over half of users have more than one year of EV experience, indicating a mature user base capable of providing insights on long-term performance, consistent with prior EV behaviour research [29].

Analysis of daily travel distances shows that 80% of respondents commute less than 60 km per day, which is sufficient to meet typical daily mobility needs, consistent with Tong and Feng [30]. Charging frequency patterns reveal that users charge once daily, and once every 2–3 days, aligning with the charging behaviours documented by Zhang and Chen [31]. Notably, users rely primarily on public charging stations, highlighting infrastructure gaps and the need for extensive charging networks, particularly in densely populated and lower-income areas, consistent with findings by De Souza and Pereira [32] and Ermagun and Tian [33].

User satisfaction with charging infrastructure presents a mixed picture; the majority expressed satisfaction, while a large number are also equally dissatisfied. This reflects challenges in early-stage EV ecosystems, which often suffer from inconsistent reliability and accessibility, as observed by Parast and Nelson [34]. Key barriers reported by users include a lack of charging stations and unreliable power supply, echoing findings by Ghosh [35] and NITI Aayog [36]. Long waiting times and slow charging were also noted as concerns, consistent with the EV user study by Mishra and Singh [37].

Regarding preferred charging locations, residential areas and commercial zones are the top choices, emphasizing that charging networks should align with routine human mobility patterns, as suggested by Lin and Wu [38] and Pani and Mishra [39]. Pricing preferences lean toward mid-range tariffs of Rs. 31–40, highlighting the importance of affordability for lower- and middle-income groups [40]. Payment modes are dominated by UPI, reflecting the rapid adoption of digital payment systems in mobility services [41]. Peak charging typically occurs in the evening, aligning with post-work residential charging demand, consistent with the findings of Ahmed and Klinger [42] and International Energy Agency (IEA) [26].

Entrepreneurial interest in EV charging stations is strong and expresses willingness to start a business, reflecting favorable perceptions of the EV sector, as discussed by Mahajan and Arora [43]. Among business owners and potential entrepreneurs, there is a clear intent to establish micro EV charging ventures [44]. Preferred business models include Own & Operate and Shared Revenue, aligning with findings by Kumar and Khandelwal [45]. Investment preferences ranging from Rs. 50,000 to 200,000 highlight the emphasis on small-scale, manageable charging infrastructure, consistent with insights from McKinsey [46].

EV entrepreneurs also seek government support in the form of subsidies, technical training, and accessible bank loans to build capacity and enhance operational competence, as documented by UNEP [47] and the Government of India [48]. These preferences confirm that local entrepreneurs require both financial and technical backing to actively participate in EV infrastructure development [49].

Essentially, these findings highlight that EV charging should not exist in a vacuum; it needs to be woven directly into the fabric of our daily commutes. By placing charging stations at strategic “hotspots” like bus stops, markets, and offices, we can make switching between different modes of travel seamless and ensure that the “last mile” of a journey is as reliable as the first. For e-rickshaw drivers specifically, having dependable power near major transport hubs means they can keep their feeder services running without long gaps or downtime. If we then add smart time management and give priority access to these shared commercial vehicles, we can significantly cut down on congestion and wait times, making the whole city move much more efficiently. Dogan [50] recommended that policymakers in BRICS economies formulate and execute policies that promote green technology advancement, research, and development initiatives.

Collectively, these findings present a coherent picture of a rapidly evolving EV ecosystem, where consumer adoption, infrastructure demand, and entrepreneurial potential are advancing simultaneously, yet require strategic alignment and support to ensure sustainable growth.

6 Conclusion

In Nizamabad, building a reliable EV network is about much more than just installing plugs; it is about ensuring that the city’s pulse—driven by thousands of electric two-wheelers and e-rickshaws—stays consistent and connected. This study reveals that when charging stations are poorly placed or hard to find, it creates a ripple effect of long wait times and broken commutes that directly hurts the efficiency of our shared and last-mile transport. To truly make green mobility work for everyone, we have to stop looking at charging stations as mere business ventures and start treating them as a vital piece of our city’s transport DNA, just as essential as the roads themselves. Key challenges identified include insufficient stations, unreliable power supply, slow charging, and long waiting times, underscoring the importance of a well-equipped, technologically advanced charging network, particularly in residential and commercial areas. Entrepreneurial interest in EV charging businesses is notably high, with respondents favoring ownership-based or shared revenue models and moderate investment levels. Financial incentives, technical training, and accessible financing are essential to support new entrants and foster sustainable business growth. With appropriate policy and infrastructure support, Nizamabad has the potential to develop a self-sustaining EV ecosystem. Overall, the research affirms that while EV adoption is accelerating, strategic interventions targeting infrastructure expansion, enhanced user experience, and entrepreneurial capacity-building are critical to advancing a reliable and inclusive electric mobility future. Future research might look at how EV charging infrastructure fits into multimodal transportation systems, especially how it supports feeder services like shared mobility and e-rickshaws. The impact of charging availability close to public transportation hubs on connection, transfer efficiency, and overall urban mobility performance might be investigated further. The findings provide clear guidance for policymakers, urban planners, and private stakeholders to implement targeted measures that improve charging accessibility, promote innovation, and support sustainable transportation in the region.

Author Contribution

Conceptualization, B.S.K. and S.M.; methodology, B.S.K. and S.M.; software, B.S.K.; validation, B.S.K., S.M., and W.C.H.; formal analysis, B.S.K. and S.M.; investigation, S.M. and B.S.K.; resources, R.R.; data curation, B.S.K. and R.R.; writing—original draft preparation, B.S.K. and S.M.; writing—review and editing, R.R. and W.C.H.; visualization, B.S.K. and R.R.; supervision, S.M. and W.C.H.; project administration, B.S.K. and S.M.; funding acquisition, S.M. and W.C.H. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the findings of this study are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflict of interest.

References

- [1] T. Unterluggauer, J. Rich, P. B. Andersen, and S. Hashemi, “Electric vehicle charging infrastructure planning for integrated transportation and power distribution networks: A review,” *eTransportation*, vol. 12, p. 100163, 2022. <https://doi.org/10.1016/j.etrans.2022.100163>

- [2] H. R. Sayarshad, "Optimization of electric charging infrastructure: Integrated model for routing and charging coordination with power-aware operations," *npj. Sustain. Mobil. Transp.*, vol. 1, p. 4, 2024. <https://doi.org/10.1038/s44333-024-00004-6>
- [3] M. Shaat, F. Oroumchian, Z. Abohaia, and M. El Barachi, "A data-driven framework for electric vehicle charging infrastructure planning: Demand estimation, economic feasibility, and spatial equity," *World Electr. Veh. J.*, vol. 17, no. 1, p. 42, 2026. <https://doi.org/10.3390/wevj17010042>
- [4] M. I. Alhalalmeh, A. A. Sarayreh, S. I. Al-Ayed, A. A. Al-Tit, M. M. Alqahtani, M. F. A. Hunitie, T. Alanazi, S. I. S. Al-Hawary, K. M. Al-hawajreh, and A. A. S. Mohammad, "The impact of dynamic capabilities on entrepreneurial performance: An empirical study of SMEs," in *Intelligence-Driven Circular Economy*, 2025, pp. 465–479. https://doi.org/10.1007/978-3-031-74220-0_36
- [5] W. Sierzechula, S. Bakker, K. Maat, and B. Van Wee, "The influence of charging infrastructure availability on EV adoption," *Energy Policy*, vol. 120, pp. 20–35, 2018. <https://doi.org/10.1016/j.enpol.2014.01.043>
- [6] X. Li, Y. Zheng, and H. Chen, "Public charging infrastructure and EV adoption: A behavioural analysis," *Transp. Res. Part D*, vol. 67, pp. 588–604, 2019. <https://doi.org/10.1016/j.trd.2018.12.016>
- [7] H. O. Bansal and P. Goyal, "Enablers and barriers of electric vehicle in India: A review," in *2020 IEEE International Symposium on Sustainable Energy, Signal Processing and Cyber Security (iSSSC)*, Gunupur Odisha, India, 2020, pp. 1–5. <https://doi.org/10.1109/iSSSC50941.2020.9358895>
- [8] S. Jerome and M. Udayakumar, "An economic feasibility study of electric vehicle charging stations in India," in *Recent Advances in Hybrid and Electric Automotive Technologies: Select Proceedings of HEAT 2021, 2022*, pp. 207–221. https://doi.org/10.1007/978-981-19-2091-2_17
- [9] P. Jain, A. Tandon, and R. C. Bansal, "Integration of renewable energy sources (RES) into electric vehicle (EV) charging infrastructure: State-of-the-art review," in *Electric Vehicle Charging Infrastructures and its Challenges*, 2025, pp. 171–214. https://doi.org/10.1007/978-981-96-0361-9_9
- [10] R. Devanathan and M. Pillai, "Micro-entrepreneurship opportunities in India's electric mobility landscape," *J. Sustain. Bus.*, vol. 5, no. 1, pp. 22–34, 2021.
- [11] R. Jaiswal and S. Gupta, "Entrepreneurship in the electric vehicle sector: Past, present and future directions," *Parikalpana: Kiit J. Manag.*, vol. 20, no. 1, pp. 5–26, 2024. <https://doi.org/10.23862/kiit-parikalpana/2024/v20/i1/225089>
- [12] P. Saxena, R. Narayanan, and S. Dixit, "Pricing preferences and willingness to pay for EV charging in India," *Energy Policy*, vol. 179, p. 113595, 2023. <https://doi.org/10.1016/j.enpol.2023.113595>
- [13] S. Ghosh and D. Bhattacharya, "Determinants of consumer willingness to pay for EV charging services in emerging economies," *Util. Policy*, vol. 83, p. 101567, 2023. <https://doi.org/10.1016/j.jup.2023.101567>
- [14] R. S. Gupta, Y. Anand, A. Tyagi, and S. Anand, "Assessment of frameworks and initiatives for electric vehicle adoption and augmenting charging infrastructure in Indian states," *Energy Sources, Part A*, vol. 45, no. 4, pp. 11 505–11 534, 2023. <https://doi.org/10.1080/15567036.2023.2259841>
- [15] R. Kumar, A. Jha, A. Damodaran, D. Bangwal, and A. Dwivedi, "Addressing the challenges to electric vehicle adoption via sharing economy: An Indian perspective," *Manag. Environ. Qual.*, vol. 32, no. 1, pp. 82–99, 2020. <https://doi.org/10.1108/MEQ-03-2020-0058>
- [16] K. Turoń, A. Kubik, and F. Chen, "Electric shared mobility services during the pandemic: Modeling aspects of transportation," *Energies*, vol. 14, no. 9, p. 2622, 2021. <https://doi.org/10.3390/en14092622>
- [17] M. Ahmad, S. Khan, and A. Bashir, "Economic assessment of decentralized EV charging stations in urban–suburban corridors," *Energy Policy*, vol. 165, p. 112940, 2022. <https://doi.org/10.1016/j.enpol.2022.112940>
- [18] Y. Wang, Z. Li, and K. Wang, "Travel patterns and charging behaviour of EV users," *Transp. Res. Interdiscip. Perspect.*, vol. 4, p. 100094, 2020. <https://doi.org/10.1016/j.trip.2020.100094>
- [19] A. Saxena and B. Shrivastava, "Examining factors affecting the willingness of rickshaw operators to adopt battery operated rickshaws: The case of Bhopal, India," *Transp. Res. Rec.*, vol. 2677, no. 5, pp. 325–340, 2023. <https://doi.org/10.1177/03611981221130339>
- [20] M. Yilmaz and P. T. Krein, "Review of EV charging systems and implications for low-cost charging solutions," *IEEE Trans. Power Electron.*, vol. 35, no. 5, pp. 4562–4575, 2020. <https://doi.org/10.1109/TPEL.2019.2941906>
- [21] Q. Zhang, J. Hu, and Y. Ni, "Economic and operational analysis of battery swapping stations for commercial EV fleets," *Appl. Energy*, vol. 312, p. 118789, 2022. <https://doi.org/10.1016/j.apenergy.2022.118789>
- [22] A. Chauhan and R. Singh, "Determinants influencing willingness to adopt electric mobility in Indian cities," *Transp. Res. Part D*, vol. 105, pp. 103–117, 2022. <https://doi.org/10.1016/j.trd.2022.103117>
- [23] M. Ali and R. Gupta, "Consumer perception and acceptance of electric vehicles in India," *J. Clean. Prod.*, vol. 345, p. 130957, 2022. <https://doi.org/10.1016/j.jclepro.2022.130957>

- [24] P. Singh and K. Das, "User demographics and EV adoption behaviour in South Asia," *Technol. Forecast. Soc. Change*, vol. 183, p. 121896, 2022. <https://doi.org/10.1016/j.techfore.2022.121896>
- [25] S. Dutta and A. Mandal, "Affordability and operational savings of electric vehicles in low-income markets," *Energy Econ.*, vol. 98, pp. 105–112, 2021. <https://doi.org/10.1016/j.eneco.2021.105112>
- [26] International Energy Agency (IEA), "Global EV outlook 2023," 2023, OECD/IEA Publications. <https://www.iea.org/reports/global-ev-outlook-2023>
- [27] H. Breetz, M. Mildenerberger, and L. Stokes, "The political logic of clean energy transitions," *Bus. Polit.*, vol. 22, no. 4, pp. 615–639, 2020. <https://doi.org/10.1017/bap.2018.14>
- [28] S. Bhattacharya and A. Saha, "Electric vehicle infrastructure: A review of global charging patterns and user behaviour," *Renew. Sustain. Energy Rev.*, vol. 152, p. 111650, 2021. <https://doi.org/10.1016/j.rser.2021.111650>
- [29] K. Joshi and A. Verma, "Understanding range anxiety and charging behaviour among EV users," *Sustain. Energy Technol. Assess.*, vol. 51, p. 101879, 2022. <https://doi.org/10.1016/j.seta.2021.101879>
- [30] F. Tong and Y. Feng, "Daily travel distance and EV range adequacy in urban environments," *Transp. Res. Part A*, vol. 136, pp. 91–106, 2020. <https://doi.org/10.1016/j.tra.2020.03.016>
- [31] X. Zhang and G. Chen, "Charging frequency and behaviour modelling among EV users," *Appl. Energy*, vol. 312, p. 118793, 2022. <https://doi.org/10.1016/j.apenergy.2022.118793>
- [32] L. De Souza and M. Pereira, "Charging infrastructure planning for electric vehicles in urban landscapes," *Sustain. Cities Soc.*, vol. 69, p. 102881, 2021. <https://doi.org/10.1016/j.scs.2021.102881>
- [33] A. Ermagun and J. Tian, "Charging into inequality: A national study of social, economic, and environment correlates of electric vehicle charging stations," *Energy Res. Soc. Sci.*, vol. 115, p. 103622, 2024. <https://doi.org/10.1016/j.erss.2024.103622>
- [34] M. Parast and R. Nelson, "Infrastructure reliability and user satisfaction in EV charging," *Transp. Res. Part F*, vol. 95, pp. 40–53, 2023. <https://doi.org/10.1016/j.trf.2023.03.002>
- [35] A. Ghosh, "Electric vehicle charging station infrastructure: A critical review," *Environ. Sci. Pollut. Res.*, vol. 27, no. 31, pp. 38 775–38 793, 2020. <https://doi.org/10.1007/s11356-020-10131-3>
- [36] NITI Aayog, "EV Charging Infrastructure Standards and Guidelines for India," 2022. https://www.niti.gov.in/sites/default/files/2023-02/EV_Handbook_Final_14Oct.pdf
- [37] A. Mishra and S. Singh, "Public charging infrastructure and user satisfaction metrics," *Int. J. Energy Res.*, vol. 46, no. 12, pp. 15 544–15 559, 2022. <https://doi.org/10.1002/er.8175>
- [38] J. Lin and F. Wu, "Optimizing EV charging locations for mixed residential-commercial zones," *J. Urban Plann. Dev.*, vol. 147, no. 3, 2021. [https://doi.org/10.1061/\(ASCE\)UP.1943-5444.0000715](https://doi.org/10.1061/(ASCE)UP.1943-5444.0000715)
- [39] A. Pani and P. Mishra, "Spatial planning of EV charging stations using GIS," *J. Transp. Land Use*, vol. 14, no. 1, pp. 211–231, 2021. <https://doi.org/10.5198/jtlu.2021.1802>
- [40] L. Eanes and J. Brazell, "Consumer willingness to pay for public EV charging convenience," *J. Transp. Geogr.*, vol. 108, p. 103580, 2023. <https://doi.org/10.1016/j.jtrangeo.2023.103580>
- [41] V. Jain and R. Sharma, "Impact of digital payment systems on EV charging adoption," *J. Retail. Consum. Serv.*, vol. 62, p. 102630, 2021. <https://doi.org/10.1016/j.jretconser.2021.102630>
- [42] S. Ahmed and J. Klinger, "A pricing strategy for electric vehicle charging in residential areas considering the uncertainty of charging time and demand," *Comput. Commun.*, vol. 199, pp. 153–167, 2023. <https://doi.org/10.1016/j.comcom.2022.12.018>
- [43] S. Mahajan and T. Arora, "Entrepreneurial opportunities in India's EV market," *J. Entrep. Emerg. Econ.*, vol. 15, no. 1, pp. 45–66, 2023. <https://doi.org/10.1108/JEEE-05-2022-0167>
- [44] R. Pal and S. Bose, "The economic feasibility of micro-entrepreneurship in electric vehicle charging," *Energy Sustain. Dev.*, vol. 65, pp. 87–96, 2021. <https://doi.org/10.1016/j.esd.2021.08.003>
- [45] A. Kumar and R. Khandelwal, "Analysis of slow and fast charging models for urban EV deployment," *Energy Storage*, vol. 7, no. 2, p. 104312, 2023. <https://doi.org/10.1002/est2.4312>
- [46] K. Heineke, N. Laverty, T. Möller, and F. Ziegler, "The future of mobility," 2023. <https://www.mckinsey.com/~media/mckinsey/industries/automotive%20and%20assembly/our%20insights/the%20future%20of%20mobility/the-future-of-mobility.pdf>
- [47] United Nations Environment Programme, "Progress in the implementation of resolution 4/3 on sustainable mobility: Report of the executive director," UNEP/EA.6/3. <https://docs.un.org/en/UNEP/EA.6/3>
- [48] Ministry of Power and Government of India, "Guidelines for Installation and Operation of Electric Vehicle Charging Infrastructure-2024 reg." 2023, No. 12/2/2018-EV (Comp No. 241852). https://powermin.gov.in/sites/default/files/Guidelines_and_Standards_for_EVCI_dated_17.09_2024.pdf
- [49] Y. Liu and M. Wei, "Battery swapping technology: Adoption potential and operational challenges," *Energy Rep.*, vol. 8, pp. 2192–2205, 2022. <https://doi.org/10.1016/j.egyr.2022.01.139>

- [50] E. Doğan, “Green innovation and total factor productivity relationship: The case of Türkiye,” *Ekonomika*, vol. 104, no. 4, pp. 78–94, 2025. <http://doi.org/10.15388/ekon.2025.104.4.5>